UNIVERSITY OF ESSEX

Postgraduate Examinations 2018

MACHINE LEARNING AND DATA MINING

Time allowed: **TWO** hours

Candidates are permitted to bring into the examination room:
Calculator – Casio FX-83GT PLUS or Casio FX-85GT PLUS only

The paper consists of **THREE** questions.

Candidates must answer **ALL** questions.

The questions are **not** of equal weight.

Questions 1 and 2 are worth 35% of the marks for the paper each.
Question 3 is worth 30% of the marks for the paper.

The percentages shown in brackets provide an indication of the proportion of the total marks for the **PAPER** that will be allocated for that part of the question.

Please do not leave your seat unless you are given permission by an invigilator.
Do not communicate in any way with any other candidate in the examination room.
Do not open the question paper until told to do so.
All answers must be written in the answer book(s) provided.
All rough work must be written in the answer book(s) provided. A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be marked.
At the end of the examination, remain seated until your answer book(s) have been collected and you have been told you may leave.

Question 1

(a) The following data are used to train a system to predict sales performance of a set of objects produced by a company: [10%]

Colour	*Type*	*Material*	*Sales Increase*
White	Indoor	Wood	Yes
White	Indoor	Wood	Yes
Green	Indoor	Plastic	Yes
Green	Indoor	Wood	No
White	Indoor	Plastic	No
White	Indoor	Wood	Yes
White	Indoor	Plastic	No
White	Indoor	Plastic	No
Green	Indoor	Plastic	Yes
Green	Outdoor	Plastic	Yes
Green	Outdoor	Wood	No
Green	Outdoor	Wood	No

Calculate the information gain provided by each of the three attributes: Colour, Type and Material.

(You may find it helpful to remember that $\log_2(x) = \ln(x)/\ln(2) = \log_{10}(x)/\log_{10}(2)$.)

(b) Starting from the results of part (a) of this question, construct the complete decision tree to predict the sales performance of new objects. Show the calculations or reasoning behind each step. [13%]

(c) Standard “greedy” decision tree induction algorithms select features by taking into consideration the immediate contribution of a candidate attribute at the current step and ignoring its potential contribution to obtain an efficient classification tree in future steps, for example in combination with other attributes. Discuss how a shallower tree could be obtained in part (b) of this question by selecting an initial split with a lower information gain. Then design a modified decision tree induction algorithm that is able to account for and exploit interactions between pairs of attributes. Finally, discuss a possible limitation on the applicability of your proposed algorithm. [12%]

Question 2

(a) Explain what is meant by the term Markov Decision Process (MDP) and explain the meaning of the elements of the tuple used to formally define a MDP. [10%]

(b) Explain the following equation in the context of reinforcement learning: [6%]

$$V^*(s) = \max_{a \in A(s)} [R(s,a) + \gamma V^*(s')]$$

(c) Consider a reinforcement learning problem modelled as a MDP with deterministic transitions and actions. The states are S = {A, B, C, D, E, G1, G2, G3} while the actions are trivially A = {toA, toB, toC, toD, toE, toG1, toG2, toG3}. The possible transitions and the corresponding rewards (if any) are indicated in the state transition diagram shown in Figure 2.1 below. Assuming a discount factor 0.5, calculate the discounted cumulative value of each state, also providing a brief explanation of the procedure followed. [13%]

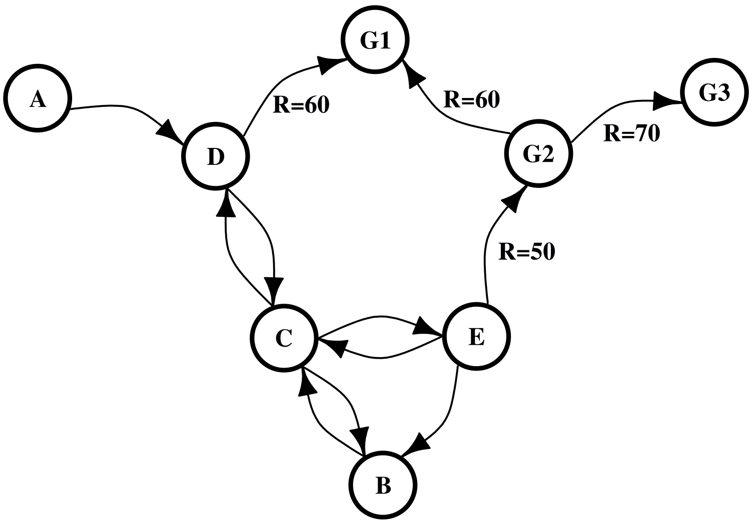

Figure 2.1

(d) Discuss the role of the discount factor in the context of reinforcement learning. In particular, consider your answer to part (c) of this question and discuss how the optimal policy from a given state, for example A, changes depending on the choice of the discount factor γ. [6%]

Question 3

(a) Explain what is meant by each of the following terms in the context of association rule mining and write the corresponding probabilistic definition: [7%]

(i) Confidence

(ii) Support

(iii) Lift

(b) An online store is using association rules as the basis for decisions about their marketing activities, product placement and promotional pricing. The first nine transactions recorded are as follows: [12%]

Transaction 1: A, B
Transaction 2: C, D
Transaction 3: A, C, E
Transaction 4: B, C
Transaction 5: A, C, D
Transaction 6: A, B
Transaction 7: A, B, C, E, F
Transaction 8: B, C
Transaction 9: A, B, C, G

By applying the Apriori Algorithm, find all the frequent itemsets for this data using a minimum support threshold of 20%. For each candidate itemset which is not retained, specify whether it was discarded because of a non-frequent subset (Apriori property) or because of insufficient support.

(c) Assuming a confidence threshold 75%, find all strong association rules of the form {X,Y}=>{Z}, state their confidence, and calculate their lift. [6%]

(d) Discuss the computational cost of the Apriori algorithm compared, for example, to a brute force approach that tries all possible itemsets. Examine which aspects of the Apriori algorithm make this computational saving possible. (Suggestion: consider your answer to part (c) of this question and compare the number of times the support of an itemset was required by the Apriori algorithm with an estimate of the number that would have been required if trying all possible itemsets.) [5%]

END OF PAPER CE802-7-AU